GN_2 Pressurant Line Sizing: McMaster-Carr Seamless Stainless Tubing Above and Below the Aqua 873-D Regulator

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Abstract

Seamless 304 and 316 stainless tubing from McMaster-Carr in 1/4, 3/8 and 1/2 in OD (walls 0.020–0.083 in) was checked for the GN_2 line between the Luxfer T95A bottle and the regulator (upstream) and for the lines from the regulator to the LOx and IPA tanks (downstream). The checks cover a nominal firing and a full-flow regulator failure. The criteria are a safety factor of at least 1.5 on yield and 3.5 on burst at the maximum expected operating pressure (MEOP), using the minimum wall allowed by ASTM A269, and a Mach number below 0.3 everywhere in the tube. **Upstream:** 1/4 in × 0.035 in passes (SF_y 1.80, SF_b 5.10 at 3500 psig; Mach 0.20 nominal and 0.21 in the failure case). **Downstream:** every 1/2 in wall passes. 1/2 in × 0.049 in is recommended (relief-limited MEOP 644 psig, SF_y 7.5, Mach 0.10 nominal and 0.20 in the failure case). 1/4 in tubing cannot carry the nominal downstream demand. 3/8 in passes only with walls of 0.028 in or less. The 0.125 in orifice ahead of the regulator is what makes the relief valves adequate. With it, the two relief valves can pass 2.69 times the failure flow. Without it, the margin drops to 1.02.

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1 Scope, inputs and answers to the brief

System. The system, as modelled in the earlier regulator study [1], is shown in Figure 1. A Luxfer T95A Type 3 COPV (13.1 L water volume, 3500 psi service pressure [2]) feeds a 0.125 in orifice at the regulator inlet. The regulator is an Aqua 873-D, dome-loaded and set to 500 psig lock-up, with $C_v = 0.8$ [3]. A common tube runs to a tee. Each branch then passes a check valve ($C_v = 1.9$) and

ends in the pintle diffuser (8 holes of 5/32 in) of its tank. Both propellant tanks are now NPS 8 Schedule 10S pipe 10 in long with two 2:1 ellipsoidal heads and 23 % initial ullage. Each tank has a relief valve cracking at 550 psig, with $K_d = 0.35$ and a 0.515 in orifice.

COPV 3500 psig $\xrightarrow{1.0\text{ m},\ K=3}$ **[UPSTREAM TUBE]** → 0.125 in orifice → 873-D (500 psig) $\xrightarrow{0.5\text{ m},\ K=0.5}$ tee

tee $\xrightarrow{K=1}$ **[BRANCH TUBE 1.5 m,** $K = 3$**]** → check valve C_v 1.9 → 0.3 m → diffuser → LOx tank (relief 550 psig)

tee $\xrightarrow{K=1}$ **[BRANCH TUBE 1.5 m,** $K = 3$**]** → check valve C_v 1.9 → 0.3 m → diffuser → IPA tank (relief 550 psig)

Figure 1: Line layout used in the model. All three downstream tube segments use the tube under test. Lengths and fitting losses are assumptions carried over from [1].

Why line lengths are needed. The Mach number at any point is $M = \dot{m}/(\rho a A)$, so it depends on the local pressure through ρ. Friction and fitting losses, which scale with length and with the K values, set how far the pressure falls along a line. The lowest pressure, and so the highest Mach number, is at the downstream end. In the nominal case the lengths are a second-order effect for 1/2 in tube (about 15–20 psi of loss), but they decide whether 1/4 in or thick-walled 3/8 in tube can deliver the flow at all. In the failure case the losses set how far the regulator-outlet pressure rises above the relief pressure, and that pressure is the downstream MEOP. The lengths are inputs in the script and the workbook and should be replaced by the real routing.

Assumptions (each is flagged in the workbook, `Inputs` sheet, yellow cells):

1. Line lengths and lumped K values as in Figure 1. The check valve has $C_v = 1.9$ for every tube size, and the orifice C_d is 0.65.
2. Both tanks are identical. IPA density is 786 kg/m^3. Collapse factors are 1.3 for LOx (from the CF study) and 1.05 for IPA.
3. Upstream MEOP is the bottle service pressure of 3500 psig. A hot-fill value of 3850 psig is reported as a sensitivity only.
4. The failure case uses a bottle at 3500 psig and 294 K (a settled fill gives the densest gas and the largest flow). The relief gas is taken at 316 K, the warm case with the least relief capacity. The relief valves are taken as fully open at cracking pressure, with no accumulation allowance.
5. The 550 psig cracking pressure is read as gauge pressure.
6. Tube strength uses the room-temperature ASTM minimums: 30 ksi yield and 75 ksi tensile for both TP304 and TP316 [4]. Cold gas only raises strength.
7. Fitting pressure ratings and the effect of tube bending or flaring on wall thickness are not included.

Tank update. With the 10 in pipe the tank volume is 14.41 L, the initial ullage is 3.31 L and the propellant volume is 11.09 L (`Tanks` sheet). This loads 12.66 kg of LOx and 8.72 kg of IPA. At the unchanged flows of 1.243 and 0.957 kg/s, the LOx tank empties in 10.18 s but the **IPA tank empties first, at 9.12 s**. With equal tank volumes the loads do not match O/F = 1.3, which is worth checking in the fill plan. The model runs each tank until it is empty, which gives the most GN_2 use and the lowest bottle pressure.

2 Method

Section numbers in brackets refer to the `# %%` cells of `gn2_line_sizing.py`.

2.1 Structural rating [Tube candidates and structural rating]

The minimum wall comes from the ASTM A269 tolerance [5]: ±15 % for OD below 1/2 in and ±10 % for OD from 1/2 to 1-1/2 in.

$$t_{min} = t\,(1-\delta), \qquad r_o = \tfrac{OD}{2}, \qquad r_i = r_o - t_{min}. \tag{1}$$

First yield at the bore follows from the Lamé thick-wall stresses and the Tresca criterion, $\sigma_\theta - \sigma_r = 2Pr_o^2/(r_o^2 - r_i^2) = S_y$. Burst uses Barlow's formula on the OD with the minimum wall:

$$P_y = S_y\,\frac{r_o^2 - r_i^2}{2r_o^2}, \qquad P_b = \frac{2\,S_u\,t_{min}}{OD}, \qquad \mathrm{SF}_y = \frac{P_y}{\mathrm{MEOP}} \geq 1.5, \quad \mathrm{SF}_b = \frac{P_b}{\mathrm{MEOP}} \geq 3.5. \tag{2}$$

Both choices are conservative. Tresca at the bore gives a lower pressure than von Mises, and Barlow on the OD gives a lower burst pressure than the mean-diameter form.

The MEOP values are:

- upstream: 3500 psig;
- downstream: the relief-limited value, $\max(P^{fail}_{reg,out},\, P_{lock})$. Here $P^{fail}_{reg,out}$ is the regulator-outlet pressure in the failure case and $P_{lock} = 514$ psig is the highest lock-up pressure during the run (500 psig plus 3 psi per 1000 psi of inlet drop [3]).

2.2 Real-gas nozzle flow [Real-gas N2 property helpers]

All N_2 properties come from CoolProp [6]. Orifices, valves and relief valves are treated as isentropic nozzles from a stagnation state (P_0, h_0, s_0):

$$G(P) = \rho(P, s_0)\sqrt{2\,[h_0 - h(P, s_0)]}, \qquad \dot m = \begin{cases} C_d A\, G(P_d), & P_d > P^* \\ C_d A\, \max_P G = C_d A\, G^*, & \text{otherwise (choked).} \end{cases} \tag{3}$$

The critical pressure P^* is found numerically. It is $0.475P_0$ at 3500 psig and $0.53P_0$ near 1 atm. For a valve, $C_dA = 1.699 \times 10^{-5}\,C_v$ m^2, which comes from $K_v = 0.865\,C_v$ [7] and equals the usual rule $A[\mathrm{in}^2] = C_v/38$. In the ideal-gas limit Eq. (3) reduces to the API 520 critical-flow equation [8]. The workbook (`Inputs`, hand checks) evaluates that ideal-gas form with the real compressibility factor Z. It matches the real-gas relief capacity to 1 % (0.409 against 0.414 kg/s). It under-predicts the dense-gas orifice flow at 3500 psig by 10 % (0.277 against 0.307 kg/s), which is why the real-gas form is used.

2.3 Adiabatic tube flow and Mach number [1-D adiabatic tube flow]

Each tube is marched in 25 steps. The stagnation enthalpy h_0 is constant (adiabatic) and the mass flux is $G = \dot m/A$. At each step the local state is found by iterating

$$v = G/\rho, \qquad h = h_0 - v^2/2, \qquad \rho = \rho(P, h), \qquad a = a(P, h), \qquad M = v/a, \tag{4}$$

and the momentum balance with friction and fitting losses gives

$$P_{j+1} = P_j - \Big(f\frac{\Delta x}{D} + \Delta K\Big)\frac{G^2}{2\bar\rho} - G^2\Big(\frac{1}{\rho_{j+1}} - \frac{1}{\rho_j}\Big), \tag{5}$$

with the Haaland friction factor [9] and a roughness of 1.5 µm. A line is reported as *choked* ($M = 1$) when M reaches 0.99 or the march cannot pass the flow. The script marches downstream from a known inlet pressure in the nominal case and upstream from the tank pressure in the failure and pre-press cases. The two directions agree to 0.002 psi (sanity check in the script). The workbook recomputes each reported Mach number as $M = \dot m/(\rho a A)$ from the state at the worst point (`Mach` sheet).

2.4 Regulator, orifice and bottle [Regulator and bottle models, CASE A]

In the nominal case the orifice drop is found from Eq. (3). The regulator outlet then follows the droop law of [1]:

$$P_{out} = P_{lock}(P_m)\Big[1 - 0.10\,\big(\dot{m}/\dot{m}_{max}(P_m)\big)^2\Big], \tag{6}$$

where P_m is the regulator inlet pressure behind the orifice. The rated flow $\dot{m}_{max}$ is 0.221 kg/s above 1500 psig inlet and falls to zero at 500 psig. GN_2 demand per tank is $\dot{m}_i = \mathrm{CF}_i\,\rho(P_{tank}, T_{in}(t))\,Q_{prop,i}$, using the $T_{in}(t)$ history of the CF brief [10]. The bottle blows down isentropically. This is conservative for Mach number: at a given remaining mass a colder bottle has a lower pressure, so ρa is lower and M is higher. Before the run the bottle supplies the pre-pressurisation gas for both ullages (0.251 kg).

2.5 Regulator failure, relief-limited MEOP [Downstream network, CASE C]

With the poppet stuck fully open, the orifice ($C_dA = 5.15 \times 10^{-6}\,\mathrm{m}^2$) and the regulator ($C_dA = 1.36 \times 10^{-5}\,\mathrm{m}^2$) act in series. The flow $\dot{m}$ is the root of

$$C_dA_{or}\,G\big(P_b \to P_m(\dot{m})\big) = \dot{m}, \tag{7}$$

where $P_m(\dot{m})$ is the regulator inlet pressure needed to push $\dot{m}$ through the regulator into $P_{out}(\dot{m})$. $P_{out}(\dot{m})$ is in turn found by marching upstream from the tanks through the diffuser, the tube after the check valve, the check valve, the branch and the common tube. Each tank sits at the relief-limited pressure

$$P_{tank} = \max\big(P_{crack},\; P : K_dA_{rv}\,G^*(P, T_{rv}) = \dot{m}/2\big). \tag{8}$$

When the orifice is choked the solution is $\dot{m} = 0.307$ kg/s. One relief valve passes 0.414 kg/s at cracking pressure, so the two together pass 2.69 times the failure flow and the tanks stay at 550 psig. The highest downstream pressure is at the regulator outlet, and this pressure is the downstream MEOP. The upstream tube is checked at the full orifice-choked flow for every size, whatever downstream tube is fitted.

Cases. The acceptance cases are **A** (nominal firing) and **C** (regulator failure). Two further cases are information only. **B** is the regulator rated flow of 0.221 kg/s at full bottle, an upper bound on start transients. **D** is pre-pressurisation with the regulator wide open while the ullages are still at low pressure.

3 Results

3.1 Nominal run

Figure 2 shows the run. The bottle falls from 3500 psig to about 2960 psig during pre-pressurisation and to 1,231 psig and 233 K at the end of the run. It still holds 1.73 kg of its 3.08 kg. The end temperature of 233 K matches the 224 K end value of the measured T_{in} history well, which supports the isentropic-blowdown assumption. Peak demand is 0.128 kg/s, just before the IPA tank empties. Late in the run the regulator inlet behind the 0.125 in orifice falls to 834 psig. The regulator then becomes inlet-limited and the tanks dip to 444 psig. This dip comes from the regulator and orifice, not from the tubing, and was already noted in the regulator study.

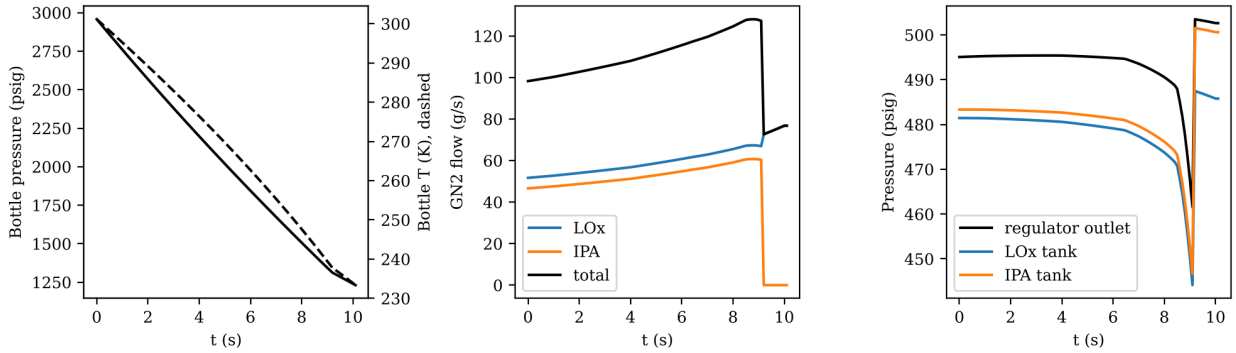


Figure 2: Case A with the 1/2 in × 0.035 in downstream reference tube. Left: bottle pressure and temperature. Centre: GN_2 flow. Right: regulator-outlet and tank pressures.

3.2 Upstream line (bottle to regulator)

304 and 316 have the same minimum strengths, so they give the same results. Table 1 lists 316 only; the `Tubes` sheet lists both. M_A is the highest Mach number over the nominal run, which occurs at the end, when the bottle pressure is lowest. M_C is the failure case at full bottle pressure.

Table 1: Upstream line: structural margin at 3500 psig and Mach number. Red values fail a criterion.

OD (in)	wall (in)	t_{min} (in)	P_y (psi)	P_b (psi)	SF_y	SF_b	M_A	M_C	verdict
1/4	0.020	0.0170	3,803	10,200	1.09	2.91	0.134	0.133	fail
1/4	0.028	0.0238	5,168	14,280	1.48	4.08	0.163	0.165	fail
1/4	0.035	0.0297	6,290	17,850	1.80	5.10	0.202	0.208	**PASS**
1/4	0.049	0.0416	8,331	24,990	2.38	7.14	0.455	0.680	fail
1/4	0.065	0.0553	10,330	33,150	2.95	9.47	choked	choked	fail
1/4	0.083	0.0706	12,154	42,330	3.47	12.09	choked	choked	fail
3/8	0.020	0.0170	2,597	6,800	0.74	1.94	0.049	0.048	fail
3/8	0.028	0.0238	3,566	9,520	1.02	2.72	0.054	0.053	fail
3/8	0.035	0.0297	4,382	11,900	1.25	3.40	0.059	0.058	fail
3/8	0.049	0.0416	5,924	16,660	1.69	4.76	0.072	0.071	**PASS**
3/8	0.065	0.0553	7,538	22,100	2.15	6.31	0.094	0.093	**PASS**
3/8	0.083	0.0706	9,164	28,220	2.62	8.06	0.135	0.135	**PASS**
1/2	0.020	0.0180	2,082	5,400	0.59	1.54	0.026	0.025	fail
1/2	0.028	0.0252	2,872	7,560	0.82	2.16	0.028	0.027	fail
1/2	0.035	0.0315	3,542	9,450	1.01	2.70	0.029	0.029	fail
1/2	0.049	0.0441	4,825	13,230	1.38	3.78	0.034	0.033	fail
1/2	0.065	0.0585	6,199	17,550	1.77	5.01	0.040	0.039	**PASS**
1/2	0.083	0.0747	7,625	22,410	2.18	6.40	0.049	0.048	**PASS**

Only 1/4 in walls of 0.035 in or less keep the Mach number below 0.3, because thicker walls leave too small a bore. Only walls of 0.035 in or more meet $SF_y \geq 1.5$. So 1/4 in × 0.035 in is the single 1/4 in tube that passes. Its yield SF at the 3850 psig hot-fill pressure is 1.63. 3/8 in passes from a 0.049 in wall and 1/2 in from 0.065 in. Either has much lower Mach numbers (0.04–0.14), but needs adapters at the 1/4 in NPT regulator inlet.

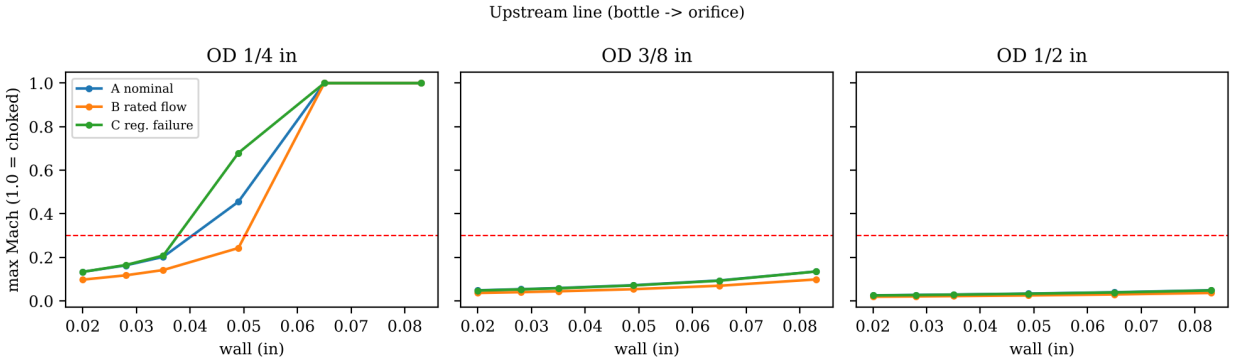


Figure 3: Upstream Mach number against wall thickness. A value of 1.0 means the line chokes.

3.3 Downstream lines (regulator to tanks)

Table 2: Downstream lines: relief-limited MEOP, safety factors, Mach number in cases A and C, case B (information), and the lowest tank pressure in the nominal run.

OD (in)	wall (in)	MEOP (psig)	SF_y	SF_b	M_A	M_C	M_B (info)	$P_{t,min}$ (psig)	verdict
1/4	0.020	1,374	2.77	7.43	choked	0.405	choked	–	fail
1/4	0.028	1,609	3.21	8.87	choked	0.458	choked	–	fail
1/4	0.035	1,857	3.39	9.61	choked	0.517	choked	–	fail
1/4	0.049	2,433	3.42	10.27	choked	0.631	choked	–	fail
1/4	0.065	3,042	3.40	10.90	choked	0.708	choked	–	fail
1/4	0.083	3,408	3.57	12.42	choked	0.678	choked	–	fail
3/8	0.020	711	3.65	9.56	0.149	0.264	0.339	432	**PASS**
3/8	0.028	740	4.82	12.87	0.165	0.284	0.388	424	**PASS**
3/8	0.035	771	5.68	15.43	0.182	0.302	choked	416	fail
3/8	0.049	861	6.88	19.35	0.225	0.339	choked	389	fail
3/8	0.065	1,032	7.30	21.41	0.303	0.377	choked	321	fail
3/8	0.083	1,387	6.61	20.35	choked	0.406	choked	–	fail
1/2	0.020	620	3.36	8.71	0.078	0.153	0.165	453	**PASS**
1/2	0.028	625	4.60	12.10	0.084	0.164	0.178	451	**PASS**
1/2	0.035	630	5.62	15.00	0.089	0.174	0.191	450	**PASS**
1/2	0.049	644	7.49	20.54	0.102	0.196	0.220	447	**PASS**
1/2	0.065	668	9.28	26.26	0.121	0.226	0.266	442	**PASS**
1/2	0.083	713	10.70	31.45	0.150	0.266	0.342	431	**PASS**

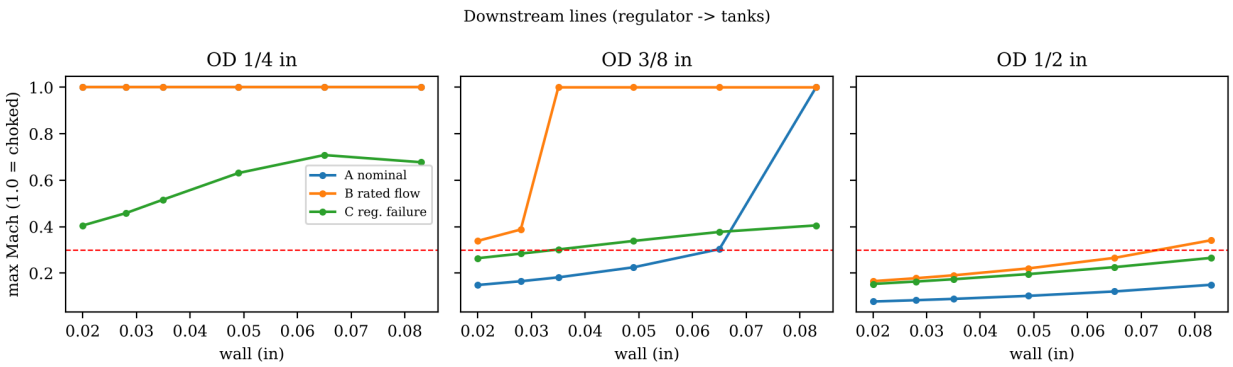


Figure 4: Downstream Mach number against wall thickness. A value of 1.0 means the line chokes.

Structural margin is large for every downstream tube, because the relief-limited MEOP is only 620–1400 psig. The Mach limit decides the size:

- **1/4 in** chokes at nominal demand. Tank pressure would collapse to below about 250 psig.
- **3/8 in** passes case C only with walls of 0.028 in or less (Mach 0.26–0.28). The common tube from the regulator to the tee, which carries both flows, limits it. These walls are thin for compression-fitting gas seals. At 0.035 in, Mach is 0.302. 3/8 in × 0.065 in also loses more than 100 psi in nominal operation.
- **1/2 in** passes at every wall. For 0.049 in: MEOP 644 psig, SF_y 7.5, SF_b 20.5, Mach 0.10 nominal and 0.20 in failure. It also matches the 1/2 in NPT regulator outlet.

3.4 Failure case details and sensitivities

Table 3: Case C for 3/8 in and 1/2 in tube, with two sensitivities: only one relief valve opens (all flow through one branch), and the 0.125 in orifice is absent.

OD (in)	wall (in)	$\dot{m}_C$ (kg/s)	$P_{reg,in}$ (psig)	$P_{reg,out}$ (psig)	M_C	M, 1 relief	P_{out}, 1 relief (psig)	M, no orifice
3/8	0.020	0.307	1,313	711	0.264	0.305	967	0.395
3/8	0.028	0.307	1,318	740	0.284	0.336	1,025	0.428
3/8	0.035	0.307	1,326	771	0.302	0.366	1,088	0.466
3/8	0.049	0.307	1,358	861	0.339	0.441	1,262	0.554
3/8	0.065	0.307	1,449	1,032	0.377	0.554	1,571	0.658
3/8	0.083	0.307	1,703	1,387	0.406	0.705	2,073	0.759
1/2	0.020	0.307	1,310	620	0.153	0.163	772	0.272
1/2	0.028	0.307	1,310	625	0.164	0.175	784	0.287
1/2	0.035	0.307	1,310	630	0.174	0.186	796	0.300
1/2	0.049	0.307	1,310	644	0.196	0.213	826	0.328
1/2	0.065	0.307	1,310	668	0.226	0.251	878	0.361
1/2	0.083	0.307	1,313	713	0.266	0.307	971	0.396

- With the orifice, the failure flow of 0.307 kg/s is fixed by the orifice. The regulator inlet then sits at about 1,310 psig and the downstream gas is at about 270 K after Joule–Thomson throttling.
- **Without the orifice**, the regulator C_v alone passes 0.812 kg/s. The two relief valves can then pass only 1.02 times the flow at cracking pressure, almost no margin. Downstream Mach rises to 0.33 for 1/2 in × 0.049 in (0.30 for 0.035 in). The orifice is therefore a safety-critical part. It should be documented as such, and it should be neither removed nor enlarged without re-running the relief and line checks.
- **One relief valve fails to open:** all the flow goes through one branch. The line pressure rises to 826 psig for 1/2 in × 0.049 in and Mach is 0.21. The structural margin stays large.
- **Pre-pressurisation (case D, Figure 5):** with the regulator wide open into low-pressure ullages, the flow is compressible in every tube size at first. For 1/2 in × 0.049 in, Mach falls below 0.3 once the tanks pass about 300 psig. Pre-pressurisation is not a firing condition. If incompressible treatment matters there too, use a slower, throttled fill.

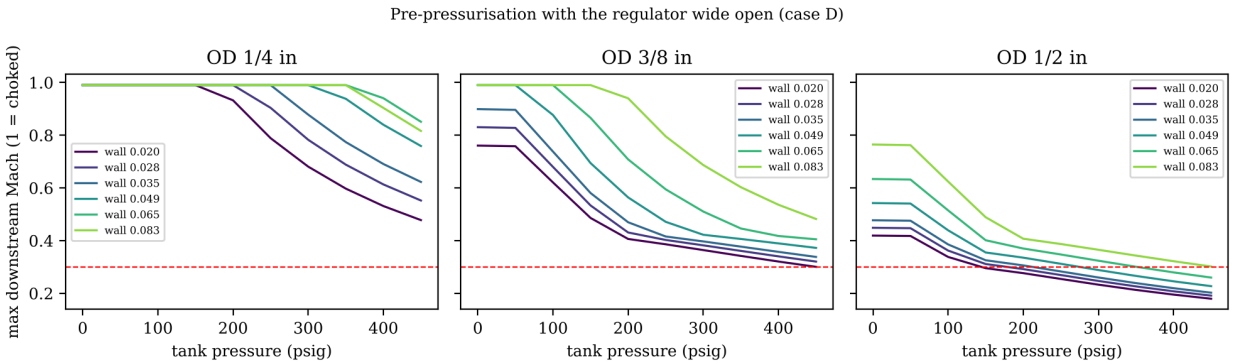


Figure 5: Case D: highest downstream Mach number against tank pressure during pre-pressurisation.

4 Recommendation

Table 4: Recommended tubing (McMaster-Carr Smooth-Bore Seamless, ASTM A213/A269, annealed [11]).

Line	Tube	316 part (304 part)	Margins	Mach (A / C)
Upstream (bottle–regulator)	1/4 × 0.035 in	89785K824 (89895K724)	SF_y 1.80, SF_b 5.10	0.20 / 0.21
Downstream (regulator–tanks)	1/2 × 0.049 in	89785K845 (89895K745)	SF_y 7.5, SF_b 20.5	0.10 / 0.20
Downstream, lighter option	1/2 × 0.035 in	89785K844 (89895K744)	MEOP 630 psig, SF_y 5.6	0.09 / 0.17

The upstream choice has the least margin: SF_y is 1.80 against 1.5 required, and Mach is 0.20 late in the run. If the real upstream run is much longer than 1 m or has more fittings, 3/8 in × 0.049 in (89785K838) is a robust alternative. The 1/2 in × 0.049 in downstream tube is preferred to 0.035 in because the heavier wall is more robust and better suited to compression-fitting gas service. The trade-off is a slightly smaller bore: Mach 0.20 against 0.17 in case C, and 0.33 against 0.30 if the orifice were absent. Check the wall against the fitting maker's gas-service guidance before ordering. Use 316 where the tube can see oxidiser-side contamination or outdoor exposure. 304 gives the same pressure ratings.

Open items. (1) Replace the assumed line lengths and fittings with the real routing and re-run the script. (2) Confirm the check-valve C_v, which is smaller for 1/4 in valves. (3) Confirm the orifice C_d and that the orifice is permanent. (4) Confirm whether the relief cracking pressure is gauge and whether full lift needs overpressure. (5) Confirm the fill plan given that IPA empties first. (6) Check fitting and adapter pressure ratings, including 1/4 in NPT and 1/2 in NPT to tube and MS33649-16 to tube, against 3500 psig and 644 psig.

Note on McMaster ratings. McMaster lists "not rated" for many of these tubes, and its listed ratings are not monotonic with wall thickness (for example 316, 1/4 in: 7800 psi at 0.049 in but 5200 psi at 0.065 in). This report uses its own ASTM-based calculation and does not rely on the catalogue figures. The catalogue values are listed in Table 5 for reference.

Table 5: McMaster-Carr part numbers checked (read 26 Oct 2026). Verify them before ordering.

OD (in)	wall (in)	304 part	316 part	ID (in)	McMaster 316 rating (psi)
1/4	0.020	89895K722	89785K822	0.210	1,600
1/4	0.028	89895K723	89785K823	0.194	2,200
1/4	0.035	89895K724	89785K824	0.180	not rated
1/4	0.049	89895K725	89785K825	0.152	7,800
1/4	0.065	89895K726	89785K826	0.120	5,200
1/4	0.083	89895K727	89785K827	0.084	6,600
3/8	0.020	89895K735	89785K835	0.335	1,050
3/8	0.028	89895K736	89785K836	0.319	2,950
3/8	0.035	89895K737	89785K837	0.305	not rated
3/8	0.049	89895K738	89785K838	0.277	2,600
3/8	0.065	89895K739	89785K839	0.245	3,450
3/8	0.083	89895K741	89785K841	0.209	4,400
1/2	0.020	89895K742	89785K842	0.460	800
1/2	0.028	89895K743	89785K843	0.444	1,100
1/2	0.035	89895K744	89785K844	0.430	not rated
1/2	0.049	89895K745	89785K845	0.402	not rated
1/2	0.065	89895K746	89785K846	0.370	not rated
1/2	0.083	89895K747	89785K847	0.334	3,300

Files

`gn2_line_sizing.py` contains the full model, with all inputs at the top, and writes `results/`, `figures/` and `GN2_line_sizing.xlsx`. The workbook keeps tube ratings, safety factors, Mach numbers and verdicts as live formulas. The orange cells hold real-gas results from the script. `report/make_tables.py` builds the tables and numbers in this document from `results/`.

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